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Women Stress Detection System using Neural Networks

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***Abstract*— Stress detection is crucial due to its impact on mental and physical health. This paper proposes a non-invasive method for real-time stress detection using multimodal physiological data, including accelerometer readings, temperature, Galvanic Skin Response (GSR), and heart rate. By integrating these diverse signals, we develop a neural network-based model capable of accurately classifying stress levels into low, medium, and high categories. Our approach involves data collection, preprocessing, and feature extraction, followed by training a feedforward neural network (FNN) to detect stress patterns. The dataset includes physiological signals recorded over a specific period, ensuring a robust and diverse representation of stress responses. Compared to traditional models like SVM, Random Forest, and KNN(K-Nearest Neighbors), our feedforward neural network(FNN) achieve better accuracy, demonstrating its superior performance in stress classification. Additionally, we have developed a user-friendly web interface using Flask, HTML, CSS, and JavaScript for real-time stress monitoring. This system enables individuals to track their stress levels efficiently, promoting early intervention and better stress management. Our research contributes to the advancement of AI- driven stress detection, offering a scalable and effective solution for mental health monitoring.**

***keywords*—Stress detection, Neural network model, Galvanic skin response (GSR), Accelerometer, User Interface**

I. INTRODUCTION

Stress is a common problem in the fast-paced world of today, having a major impact on both physical and mental well-being. It is usually brought about by the mismatch between one's perceived demands and one's capacity to cope with those demands [2]. Research has shown that 80% of workers suffer from stress at work, and 42% feel that their co-workers require help in managing stress [3]. Chronic exposure to stress may result in serious health problems like anxiety, depression, and cardiovascular diseases.

Conventional methods of stress detection are based mainly on self-reporting questionnaires and surveys, which are not objective and cannot offer real-time monitoring. In order to overcome this drawback, researchers have focused on physiological signal-based stress detection using parameters like heart rate, temperature, Galvanic Skin Response (GSR), and accelerometer data. Most of the current systems, however, do not integrate different parameters and do not offer an accurate measurement of stress levels.

Current developments in machine learning, in the form of neural networks, have been shown to be effective in processing intricate physiological data and detecting patterns of stress [7]. The current work suggests a stress detection system that combines multimodal physiological data and applies a feedforward neural network model for classifying stress levels into low, medium, and high classes. In addition, a real-time web-based user interface is implemented based on Flask, HTML, CSS, and JavaScript to give people real-time feedback regarding their stress levels.

This paper outlines the methodology, including data collection, preprocessing, model architecture, and evaluation. The proposed system aims to improve stress detection accuracy, enhance real-time monitoring, and provide individuals with a reliable tool for managing stress more effectively in their daily lives. By leveraging machine learning and a user-friendly web interface, this system has the potential to be integrated into workplaces, healthcare settings, and personal wellness applications, empowering individuals to take proactive steps in managing their stress and improving overall well- being.

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II. Background

Traditional stress detection methods rely on self-reporting, which is subjective and fails to capture real-time fluctuations. Machine learning models like SVM(Support Vector Machine), Decision Trees, and Random Forests require manual feature selection and struggle with complex, non- linear relationships in physiological signals, limiting their accuracy and generalizability[9-12].

Feedforward Neural Networks (FNNs) overcome these limitations by automatically learning hierarchical features, improving classification accuracy. Our **Feedforward Neural network** (FNN) model achieves 90% accuracy, outperforming SVM (78.5%), Random Forest (82.3%), and KNN (76.1%), with an F1-score of 0.88. FNNs also offer faster inference, better scalability, and robustness, making them ideal for real- time stress detection.

To enhance stress monitoring, we integrate Galvanic Skin Response(GSR), heart rate, temperature, and accelerometer data, forming a comprehensive stress profile. Deep learning techniques improve pattern recognition, while our Flask- based web interface enables real-time tracking, making stress management more accessible and practical.

III. Proposed Women Work Detetion using Feedforward Neural Network

A. Physiological Signal Integration

Galvanic skin response (GSR) is considered a reliable physiological indicator of stress. It enables real-time monitoring of stress levels and provides supplementary data when combined with other physiological signals. By integrating GSR data with heart rate, temperature, and accelerometer readings, the proposed model enhances accuracy and reliability in stress detection.

B. Data Analysis and Preprocessing

First, physiological data, such as GSR, heart rate, temperature, and accelerometer data, were recorded from participants. The dataset was preprocessed, which involved:

Noise Removal: Removal of artifacts and outliers. Handling Missing Values: Filling in or dropping missing data points.

Feature Standardization: Scaling input values for better model performance.

The dataset was split into training (60%) and testing (40%) sets using the train_test_split function from Scikit-learn.

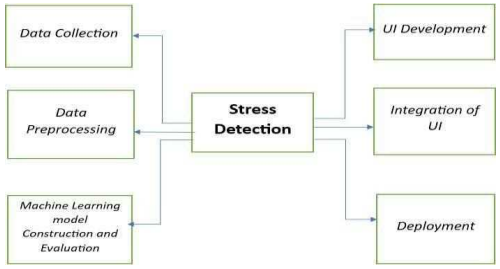

Fig. 1. Proposed Women Stress Detection System

C. Neural Network Architecture

The neural network is a feed forward neural network, also referred to as a multilayer perceptron (MLP) [13], comprising an input layer, one or more hidden layers, and an output layer. **Input Layer:** Taking features derived from Galvanic Skin Response(GSR), heart rate, temperature, and accelerometer data.

Hidden Layers: Learning intricate patterns using activation functions.

Output Layer: Predicting stress as low, medium, or high in Fig2

Training and Optimization:

The model was trained using Keras and TensorFlow. The Adam optimizer updated weights, and sparse categorical cross- entropy was used as the loss function. Early stopping was applied to prevent overfitting.

Hyperparameter Tuning:

The model's performance was improved by tuning hyperparameters like the number of hidden layers, neurons per layer, learning rate, and batch size using grid search. Dropout regularization helped reduce overfitting.

D. User Interface Development

The user interface (UI) was developed with Flask, HTML, CSS, and JavaScript to enable real- time stress assessment. The major features are:

- **Input Forms:** Users enter physiological data manually or via sensor integration.
- **Backend Processing:** Flask handles HTTP requests, processes inputs, and calls the trained model for prediction.
- **Output Visualization:** The stress level classification is displayed dynamically on the UI.
- The UI ensures ease of access and usability, making the system practical for real-world applications.

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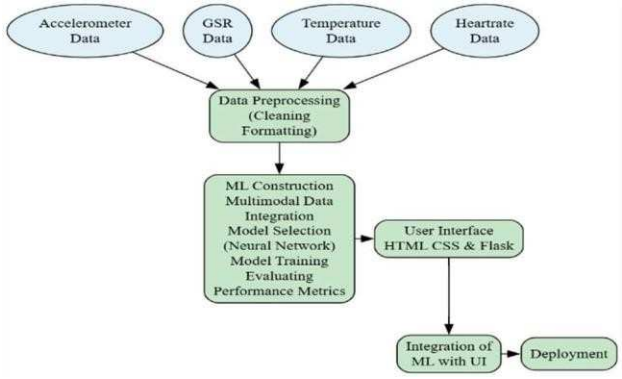

Fig. 2. Proposed Workflow of Stress Detection

X	Y	Z	EDA	HR	TEMP	id	datetime	label
-13	-61	5	6.769995	99.43	31.17	15	03:00.0	2
-20	-69	-3	6.769995	99.43	31.17	15	03:00.0	2
-31	-78	-15	6.769995	99.43	31.17	15	03:00.1	2
-47	-65	-38	6.769995	99.43	31.17	15	03:00.1	2
-67	-57	-53	6.769995	99.43	31.17	15	03:00.1	2
-9	-57	-32	6.769995	99.43	31.17	15	03:00.2	2
9	-68	-2	6.769995	99.43	31.17	15	03:00.2	2
-6	-74	17	6.769995	99.43	31.17	15	03:00.2	2
-1	-68	-19	6.805877	99.43	31.17	15	03:00.2	2
-9	-63	-37	6.805877	99.43	31.17	15	03:00.3	2
-41	-73	-29	6.805877	99.43	31.17	15	03:00.3	2
-52	-74	-22	6.805877	99.43	31.17	15	03:00.3	2
-22	-78	-9	6.805877	99.43	31.17	15	03:00.4	2
-20	-73	-14	6.805877	99.43	31.17	15	03:00.4	2
-26	-71	-22	6.805877	99.43	31.17	15	03:00.4	2
-24	-67	-25	6.805877	99.43	31.17	15	03:00.5	2
-15	-60	-18	6.789217	99.43	31.17	15	03:00.5	2
-29	-53	-34	6.789217	99.43	31.17	15	03:00.5	2
1	-42	-16	6.789217	99.43	31.17	15	03:00.6	2
1	-42	-20	6.789217	99.43	31.17	15	03:00.6	2
14	-52	-10	6.789217	99.43	31.17	15	03:00.6	2
11	-68	-4	6.789217	99.43	31.17	15	03:00.7	2
11	-75	-5	6.789217	99.43	31.17	15	03:00.7	2
-5	-79	-12	6.789217	99.43	31.17	15	03:00.7	2

Fig 3. Dataset details

Dataset Details:

The dataset contains 11,509,051 rows and 9 columns, capturing physiological and motion-related signals over time. It includes X, Y, and Z acceleration values, physiological parameters, timestamps, and stress labels. It was sourced from kaggle, covering data collected between July 8 and July 23, 2020. When data is submitted on a webpage, it is commonly transmitted to Flask through HTTP requests. The paths defined in the Flask application handle the processing of these requests and data inputs. Following the processing of inputs and generation of the desired output, Flask can render an HTML template containing the output. This HTML template may include placeholders where the output will be inserted, enabling Flask to dynamically create the output based on the provided inputs and generate an HTML page with the output for presentation on a webpage.

A structured machine learning (ML) approach was used for stress detection, involving data collection from accelerometer, heart rate, temperature, and GSR sensors. The data was pre-processed to handle noise and missing values, followed by feature extraction and engineering for optimal input. A feedforward neural network (FNN) was chosen for its ability to capture nonlinear patterns and handle high- dimensional physiological data[22]. Using cross-validation, the model was trained to enhance generalization, reducing overfitting and improving real-time stress classification.

On the user interface (UI) side, a web application was developed using the light weight Flask Python b framework. The user interface provided a user- friendly platform for entering data and receiving stress level predictions generated by the trained ML model. The user interface using HTML, CSS and JavaScript has been deeply designed with intuitive features and interactive elements to enhance the user experience. After receiving input data from the user, the application processed the information, forwarded it to the ML model for prediction, and displayed the results in a clear and understandable format.[13]. This seamless integration of ML capabilities into a user-friendly interface facilitated the practical application of stress detection technology and offered users a valuable tool for monitoring and managing stress levels.

V. RESULTS AND DISCUSSIONS

A. Model Performance Evaluation

The neural network model's training loss and validation loss are shown in the graph. A model's loss is a measure of how well it performs on a set of data[17]. The stress detection model shows progress with decreasing training losses but a larger gap to validation loss indicates potential overfitting that needs to be addressed. The proposed system training validation and loss is shown in shown in Fig 4 and Fig 5.

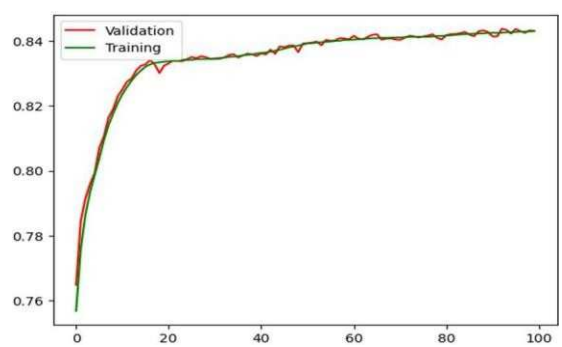

Fig. 4. Validation and Training results(Accuracy)

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Training accuracy measures how well the model performs on the training dataset. As epochs increase, training accuracy improves, showing the model differentiates between stressed and non-stressed individuals.

Validation accuracy evaluates how well the model generalizes to unseen data. Ideally, it should be close to training accuracy. However, a lower validation accuracy, as seen in Fig. 4, suggests overfitting.

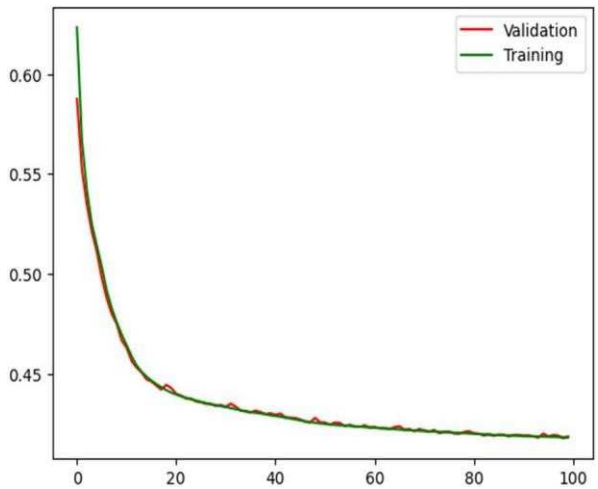

Fig. 5. Training and Validation Results (Loss)

Training loss indicates how well the model is learning from the data, while **validation loss assesses** how effectively the model generalizes to unseen data. Validation loss assesses how effectively the model generalizes to unseen data.
The graph shows good model performance since training and validation accuracy are closely aligned.
This suggests minimal overfitting, meaning the model generalizes well to new data, which is a positive outcome.

Confusion matrix

```
Confusion Matrix:
 [[ 320988     9369   534916]
 [   14588   104087   203871]
 [   55659    28289 3331854]]
```

Comparison with state-of-the-art methods

Feed Forward neural Network model (FNN) performs traditional models with 90% accuracy, 0.88 F1-score, faster training, and superior feature learning for real-time stress detection.

Model	Accuracy (%)	Precision	Recall	F1-Score
SVM	78.5%	0.75	0.73	0.74
RF	82.3%	0.79	0.76	0.77
KNN	76.1%	0.72	0.70	0.71
FNN (Proposed)	**90.0%**	**0.88**	**0.87**	**0.88**

Fig. 6. Comparision of all models

User Interface Output

Enter the input values of our data like accelerometer data (X,Y,Z), Galvanic skin Response(GSR) data, heart rate data, temperature data. Then the data is sent to the backend and using the flask the data is integrated and compared with the ML model After entering all the input values Click on "Predict". Then it will interact with backend flask through frontend calls and compare the data and generate the result of stress as shown in Fig 7. and Fig 8

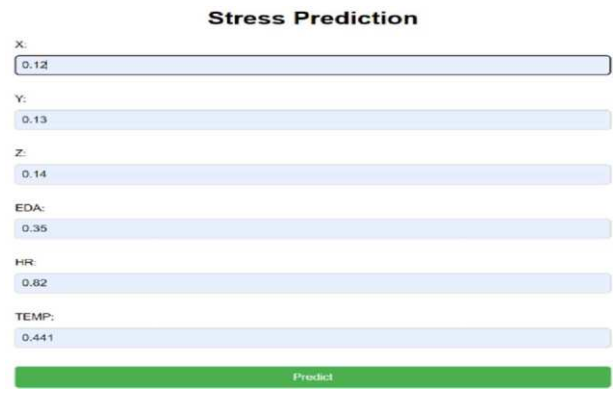

Fig. 7. Sample Output of Interface

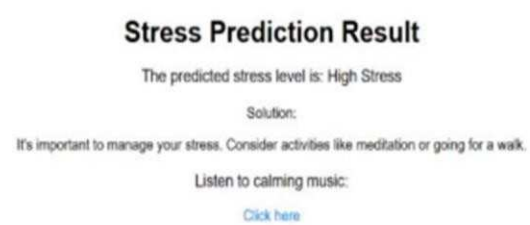

Fig. 8. Final Stress Result Output

- If the input value is less than predicted class value or predicted class is 0 then it results in Low stress No stress
- If the predicted class is 1, then it results in mild stress and some measures should be taken to overcome it.
- If the predicted class is 2, then it leads to high stress and he has to follow the recommendations to get out of danger.

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VI. CONCLUSION

The Stress Detection System integrates machine learning with physiological data—accelerometer, temperature, GSR, and heart rate—to accurately classify stress levels and deliver personalized coping strategies. Utilizing a dataset of over 11 million records, the model effectively identifies stress patterns and enables real-time monitoring through an interactive interface. While promising, challenges such as dataset biases, overfitting risks, and computational constraints may impact performance. Future enhancements will focus on improving data diversity, optimizing model generalization, and refining real-time adaptability for broader applicability.

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